

Tushar Singh

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EDUCATION

National Institute of Engineering, Mysuru

Nov 2022 - July 2026

B.E. in Computer Science and Engineering

Kendriya Vidhyalaya

May 2019 - March 2021

Senior Secondary

PROJECTS

Kidney Disease Classification – TensorFlow, Keras, DVC, AWS

- Developed a end-to-end deep learning pipeline to classify kidney CT scanned images into normal/tumour categories with 90%+ accuracy.
- Managed data versioning with DVC and experiment tracking with MLflow.
- Deployed model on AWS EC2 via FastAPI with Bootstrap frontend.
- Designed a CI/CD pipeline using GitHub Actions for automated testing and deployment.

Semantic Book Recommender – LangChain, ChromaDB, Hugging Face

- Implemented a hybrid recommendation system using Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to deliver context-aware and sentiment-driven book suggestions.
- Created vector-based semantic search system for book recommendations using OpenAI and Huggingface embeddings.
- Integrated sentiment analysis to refine recommendations by emotion tone.
- Deployed via Gradio for interactive exploration of personalised suggestions.

Azure Data Engineering Project – Azure Data Factory, Databricks, Synapse, Power BI

- Designed and implemented an end-to-end data pipeline on Azure, covering raw data ingestion, transformation, and reporting.
 - Built a multi-zone data lake architecture (Raw, Processed, Curated) using Azure Data Lake Storage Gen2.
 - Automated data workflows with Azure Data Factory and developed scalable ETL transformations with Databricks (PySpark).
 - Loaded curated datasets into Azure Synapse Analytics to support analytical queries and BI dashboards.
 - Implemented interactive Power BI dashboards to deliver actionable insights on sales and product performance.
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ACHIEVEMENTS & CERTIFICATIONS

Machine Learning

Supervised Machine Learning: Regression and Classification

Unsupervised Learning, Recommenders, Reinforcement Learning

Advanced Learning Algorithms

TECHNICAL SKILLS

Programming Languages: C++, Python

Big Data & Analytics Tools: Apache Spark, Power BI

Data Tools & Pipelines: DVC (Data Version Control), MLflow, ETL Pipelines

Databases: MySQL, MongoDB

Web & API Development: Flask, Django

Cloud Platforms: AWS, Azure

Other Tools: Github, Jupyter Labs, DagsHub, Docker